math

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description	concept
memoryless, waiting	geometric, exponential
trials with success & failure	Bernoulli, binomial
large sample, average	CLT, approximately Gaussian
bound given only mean / variance	Markov / Chebyshev inequalities
estimating parameter from data	MLE, MAP, Bayesian update
update belief from evidence	Bayes rule
function of a RV	Jensen's inequality
expected count of things	linearity + indicators
expected hitting time / return probability	first-step analysis, solve recurrence
Markov property (future depends only on present)	first-step analysis

Distributions

	discrete	continuous
counting	**binomial**: how many successes in n trials?	**Poisson**: how many events in fixed window?
waiting	**geometric**: how many trials until first success?	**exponential**: how long until first event?

Discrete

Bernoulli

- **bernoulli** RV $X \sim \text{Bernoulli}(p)$ models a single trial with p probability of success

$$P(X = 1) = p, P(X = 0) = 1 - p$$

  - compact PMF form

$$P(X = x) = p^x (1 - p)^{1-x}$$

- $\mathbb{E}[X] = p,\ \text{Var}[X] = p(1-p)$
  - proof
    - $\mathbb{E}[X] = 1 \cdot p + 0 \cdot (1-p) = p$
    - $\mathbb{E}[X^2] = 1^2 \cdot p + 0^2 \cdot (1-p) = p$
    - $\text{Var}[X] = \mathbb{E}[X^2] - \mathbb{E}[X]^2 = p - p^2 = p(1-p)$
- **categorial** RV generalizes Bernoulli from 2 to k outcomes

$$P(X = i) = p_i$$

  - each item i is $\text{Bernoulli}(p_i)$, so $\mathbb{E}[X_i] = p_i$ and $\text{Var}(X_i) = p_i(1-p_i)$

- handy techniques
  - indicator squaring: for any binary variable $X \in \{0,1\}$, $X^2 = X$ so $\mathbb{E}[X^2] = \mathbb{E}[X]$

Binomial

- **binomial** RV $X \sim \text{Binomial}(n,p)$ is the sum of n independent $\text{Bernoulli}(p)$ trials
  - $X = \sum_{i=1}^{n} X_i$ where $X_i \sim \text{Bernoulli}(p)$

$$P(X = k) = \binom{n}{k} p^k (1-p)^{n-k}$$

  - every sequence with k successes has the same probability $p^k(1-p)^{n-k}$, and there are $\binom{n}{k}$ such sequences
- **Binomial theorem**

$$(a+b)^n = \sum_{k=0}^{n} \binom{n}{k} a^k b^{n-k}$$

- we can verify that $\sum_k P(X=k) = 1$

$$\sum_{k=0}^{n} \binom{n}{k} p^k (1-p)^{n-k} = (p + (1-p))^n = 1^n = 1$$

- $\mathbb{E}[X] = np$, $\text{Var}[X] = np(1-p)$
  - proof
    - $\mathbb{E}[X] = \mathbb{E}\left[\sum_{i=1}^{n} X_i\right] = \sum_{i=1}^{n} \mathbb{E}[X_i] = np$
    - $\text{Var}[X] = \text{Var}[\sum_{i=1}^{n} X_i] = \sum_{i=1}^{n} \text{Var}[X_i] = np(1-p)$ (second equality follows because the X_i are independent)
- handy techniques
  - decompose into Bernoulli and use linearity of expectation
  - decompose X^2 into cross-terms

$$X^2 = \left(\sum_i X_i\right)^2 = \sum_i X_i^2 + \sum_{i \neq j} X_i X_j$$

    - this makes computing expectation easy (using the fact that $X_i^2 = X_i$ for indicator variables)

$$\mathbb{E}[X^2] = \sum_i \mathbb{E}[X_i] + \sum_{i \neq j} \mathbb{E}[X_i X_j] = n\mathbb{E}[X_i] + n(n-1)\mathbb{E}[X_i X_j]$$

Poisson

- **Poisson** RV $X \sim \text{Poisson}(\lambda)$ models the number of events in a fixed interval when they occur independently at an average rate of λ over that interval

$$P(X = k) = \frac{\lambda^k e^{-\lambda}}{k!}$$

- the number of events in interval of length t distributed as $N(t) \sim \text{Poisson}(\lambda t)$
- derivation using binomial limit: divide the fixed interval into n tiny time intervals, assuming it is small enough that at most one event occurs in each interval
  - then the total count is $\text{Binomial}(n, p)$ with $np = \lambda$

$$\begin{aligned} P(X = k) &= \binom{n}{k} p^k (1-p)^{n-k} \\ &= \frac{n!}{k!(n-k)!} \left(\frac{\lambda}{n}\right)^n \left(1 - \frac{\lambda}{n}\right)^{n-k} \\ &= \frac{\lambda^k}{k!} \cdot \underbrace{\frac{n!}{(n-k)!n^k}}_{\to 1} \cdot \underbrace{\left(1 - \frac{\lambda}{n}\right)^n}_{\to e^{-\lambda}} \cdot \underbrace{\left(1 - \frac{\lambda}{n}\right)^{-k}}_{\to 1} \end{aligned}$$

- $\mathbb{E}[X] = \text{Var}[X] = \lambda$
  - $\mathbb{E}[X] = \lambda$ is basically the definition of rate λ

- sums of independent Poisson are Poisson, with rates that add
  - can be derived using MGF uniqueness

$$X + Y \sim \text{Poisson}(\lambda_1 + \lambda_2)$$

Geometric

- **geometric** RV $X \sim \text{Geometric}(p)$ models the number of Bernoulli trials X until first success (inclusive), where each trial has p probability of success

$$P(X = k) = (1-p)^{k-1} \cdot p$$

  - the tail probability is a pure power of $(1-p)$

$$P(X > k) = (1-p)^k$$

- **memoryless**: $P(X > m + n \mid X > m) = P(X > n)$
  - the fact that you've failed m times already is irrelevant
  - the geometric distribution is the *only* memoryless distribution on the positive integers $\iff$ any discrete waiting-time distribution that is memoryless must be geometric
- the **geometric series** looks like $\sum_{k=0}^{\infty} r^k$
  - converges for $|r| < 1$ to $\frac{1}{1-r}$, diverges otherwise
  - can try to remember this by checking that for $r = 2$, $\sum_{k=0}^{\infty} 1/2^k \to 2$

- sanity check: PMF sums to 1

$$\begin{aligned}\sum_{k=1}^{\infty}(1-p)^{k-1}p &= p\sum_{i=0}^{\infty}(1-p)^{i} \\ &= p\cdot\frac{1}{1-(1-p)} \quad \text{sum of geometric series} \\ &= p\cdot\frac{1}{p} \\ &= 1\end{aligned}$$

- expected wait time $\mathbb{E}[X] = 1/p$
  - direct solution: $X = k$ means $k-1$ failures

$$\mathbb{E}[X] = \sum_{k=1}^{\infty} k \cdot P(X = k)$$

    - a little tricky to solve but gives $\mathbb{E}[X] = 1/p$
  - solution based on first-step analysis: on the first flip, we succeed with probability p ($X = 1$) and fail with probability $1-p$, in which case we're back to where we started ($X = 1 + \mathbb{E}[X]$)

$$\mathbb{E}[X] = p \cdot 1 + (1-p)(1 + \mathbb{E}[X])$$

    - algebra then produces $\mathbb{E}[X] = \frac{1}{p}$
  - intuition: every trial has p change of ending the process, so on average we need $1/p$ attempts
- $\mathrm{Var}(X) = \frac{1-p}{p^2}$
  - as $p \to 0$, both mean and variance blow up
  - rare successes have high variability in waiting time
- handy techniques
  - first-step analysis
  - re-index using $i = k-1$, turning $\sum_{k=1}^{\infty}$ to $\sum_{i=0}^{\infty}$

Continuous

Uniform

- **uniform** RV $X \sim \text{Uniform}(a, b)$ on $[a, b]$ means X is equally likely to be anywhere on the interval $[a, b]$
- $\mathbb{E}[X] = \frac{a+b}{2}$

$$\begin{aligned}\mathbb{E}[X] &= \int_a^b x \cdot \frac{1}{b-a} dx \\ &= \frac{1}{b-a}\frac{b^2-a^2}{2} \qquad \text{fundamental theorem of calc} \\ &= \frac{(b-a)(b+a)}{2(b-a)} \\ &= \frac{a+b}{2}\end{aligned}$$

- $\text{Var}(X) = \frac{(b-a)^2}{12}$
- PDF

$$f_X(x) = \frac{1}{b-a} \quad \text{for } a \le x \le b$$

- CDF

$$F_X(x) = \begin{cases} 0 & x < a \\ \frac{x-a}{b-a} & a \le x \le b \\ 1 & x > b \end{cases}$$

- difference between two uniform distributions is triangular
- the $n+1$ gaps between n uniform points are distributed the same way

Exponential

- **exponential** RV $X \sim \text{Exp}(\lambda)$ with rate λ models waiting time until the next event, where events happen at an average rate of λ per unit time
  - equivalently, waiting time until first event in a Poisson process with rate λ

$$f_X(x) = \lambda e^{-\lambda x}, \quad x \ge 0$$

- tail formula (aka survival function)

$$P(X > x) = e^{-\lambda x}$$

- *memoryless*: $P(X > s + t \mid X > s) = P(X > t)$
  - the fact that you've waited for s units of time already is irrelevant
  - the exponential is the *only* continuous distribution on $[0, \infty]$ that is memoryless
  - continuous analog to the geometric: waiting time in continuous time

- $\mathbb{E}[X] = \frac{1}{\lambda}$, $\text{Var}(X) = \frac{1}{\lambda^2}$
  - if events happen at rate λ/unit time on average, then on average you need to wait $1/\lambda$ units of time between events
- relation between Poisson and exponential
  - two equivalent statements
    - number of events in unit interval is $\text{Poisson}(\lambda)$
    - inter-event time is $\text{Exp}(\lambda)$
  - how many events in the next hour? → Poisson
  - how long until the next event? → exponential
- minimum of exponentials: if $X_1 \sim \text{Exp}(\lambda_1)$ and $X_2 \sim \text{Exp}(\lambda_2)$ are independent, then

$$\min(X_1, X_2) \sim \text{Exp}(\lambda_1 + \lambda_2)$$

  - intuition: if emails arrive at rate λ_1 and texts at rate λ_2, then the waiting time for "either an email or a text" is exponential with rate $\lambda_1 + \lambda_2$
  - proof

$$\begin{aligned} P(\min(X_1, X_2) > t) &= P(X_1 > t \text{ and } X_2 > t) \\ &= P(X_1 > t)P(X_2 > t) \quad \text{independence} \\ &= e^{-\lambda_1 t} e^{-\lambda_2 t} \\ &= e^{-(\lambda_1 + \lambda_2)t} \end{aligned}$$

    - which is the tail function of $\text{Exp}(\lambda_1 + \lambda_2)$

Gaussian

- **Gaussian** RV $X \sim \mathcal{N}(\mu, \sigma^2)$ with mean μ and variance σ^2
- relationship between general and standard normal RV
  - subtract the mean to shift to 0, divideå by standard dev to rescale to unit variance

$$Z = \frac{X - \mu}{\sigma} \sim N(0, 1)$$

  - example: if heights are normally distributed with $\mu = 70$ and $\sigma = 3$, what fraction of people are taller than 76 inches?

$$X \sim \mathcal{N}(70, 9) \implies Z = \frac{X - 70}{3} \sim \mathcal{N}(0, 1)$$
$$P(X > 76) = P(Z > 2)$$

- the **percentile** z_α is the value such that

$$P(Z \leq z_\alpha) = \alpha \iff P(Z > z_\alpha) = 1 - \alpha$$

  - i.e., z_α is the point on the axis with probability mass α to the *left*
  - symmetry: $z_{1-\alpha} = -z_\alpha$
  - 2-sided intervals: $P(|Z| \leq z_{1-\alpha/2}) = 1 - \alpha \iff P(|Z| > z_{1-\alpha/2}) = \alpha$
    - a $(1-\alpha)$ confidence interval uses $z_{\alpha/2}$ and $z_{1-\alpha/2}$ as endpoints
  - stated as bounds: $P(|Z| \leq c) \geq 1 - \alpha \iff c \geq z_{1-\alpha/2}$
  - this also means that for $X \sim \mathcal{N}(\mu, \sigma^2)$, the α-percentile is $x_\alpha = \mu + \sigma \cdot z_\alpha$

$$P(X \leq x_\alpha) = P\left(Z \leq \frac{x_\alpha - \mu}{\sigma}\right) = \alpha \implies z_\alpha = \frac{x_\alpha - \mu}{\sigma} \implies x_\alpha = \mu + \sigma \cdot z_\alpha$$

- sum of independent Gaussians is Gaussian: if $X_1 \sim \mathcal{N}(\mu_1, \sigma_1^2)$ and $X_2 \sim \mathcal{N}(\mu_2, \sigma_2^2)$, then

$$X_1 + X_2 \sim \mathcal{N}(\mu_1 + \mu_2, \sigma_1^2 + \sigma_2^2)$$

Expectations & variance

- **expectation**

$$\mathbb{E}[X] = \sum_x x\, P(X = k) \quad \text{or} \quad \int_{-\infty}^{\infty} x\, f(x)\, dx$$

  - expectation of a function over X

$$\mathbb{E}[g(X)] = \sum_x g(x)\, P(X = x) \quad \text{or} \quad \int_{-\infty}^{\infty} g(x)\, f(x)\, dx$$

  - **tail formula** for non-negative RVs

$$\mathbb{E}[X] = \int_0^\infty P(X > x)\, dx = \int_0^\infty 1 - F(x)\, dx$$

- **variance**

$$\begin{aligned}\mathrm{Var}(X) &= \mathbb{E}[(X - \mathbb{E}[X])^2] \\ &= \mathbb{E}[X^2] - (\mathbb{E}[X])^2\end{aligned}$$

- **linearity of expectation** (regardless of independence)

$$\mathbb{E}[aX + bY + c] = a\mathbb{E}[X] + b\mathbb{E}[Y] + c$$

- for independent X and Y

$$\mathbb{E}[XY] = \mathbb{E}[X]\mathbb{E}[Y]$$

- if X and Y are independent, then

$$\mathrm{Var}(X+Y) = \mathrm{Var}(X) + \mathrm{Var}(Y)$$

Zoo of functions

- **probability mass function** (PMF) of discrete RVs

$$p(k) = P(X = k)$$

  - properties
    - $p(k) \geq 0$
    - $\sum_k p(k) = 1$
- **probability density function** (PDF) of continuous RVs

$$f(x) \text{ such that } P(a \leq x \leq b) = \int_a^b f(x)\,dx$$

  - differentiate CDF to get PDF: $f(x) = F'(x)$

- **cumulative distribution function** (CDF) for both discrete and continuous RVs

$$F(x) = P(X \leq x)$$

  - for discrete: $F(x) = \sum_{k \leq x} p(k)$
  - for continuous: $F(x) = \int_{-\infty}^{x} f(x)\,dx$
  - note $P(a \leq x \leq b) = F(b) - F(a)$

Inequalities

- **Markov's inequality**: for any non-negative RV X and $a > 0$,

$$P(X \geq a) \leq \frac{\mathbb{E}[X]}{a}$$

  - intuitively, $\mathbb{E}[X]$ cannot be smaller than the contribution of values greater than a
  - proof

$$\mathbb{E}[X] \geq \sum_{x \geq a} xP(X = x) \geq a \sum_{x \geq a} P(X = x) = aP(X \geq a)$$

- **Chebyshev's inequality**: for any RV X with mean μ and variance σ^2

$$P(|X-\mu| \geq k\sigma) \leq \frac{1}{k^2}$$

  - at least $1 - 1/k^2$ of the distribution lies within k standard deviations of the mean
    - e.g., 75% of probability mass sits within 2 standard deviations
  - proof: apply Markov to $(X-\mu)^2$

$$P(|X-\mu| \geq k\sigma) = P((X-\mu)^2 \geq k^2\sigma^2) \leq \frac{\mathbb{E}[(X-\mu)^2]}{k^2\sigma^2} = \frac{\sigma^2}{k^2\sigma^2} = \frac{1}{k^2}$$

  - another form, by substituting $\epsilon = k\sigma$

$$P(|X-\mu| \geq \epsilon) \leq \frac{\sigma^2}{\epsilon^2}$$

- **Jensen's inequality** describes how expectations interact with convex/concave transformations

$$\begin{aligned} \varphi \text{ convex} &\implies \varphi(\mathbb{E}[X]) \leq \mathbb{E}[\varphi(X)] \\ \varphi \text{ concave} &\implies \varphi(\mathbb{E}[X]) \geq \mathbb{E}[\varphi(X)] \end{aligned}$$

  - to remember, use $\varphi(x) = x^2$ (convex) with $X \in \{-1, +1\}$
    - $\varphi(\mathbb{E}[X]) = \varphi(0) = 0$
    - $\mathbb{E}[\varphi(X)] = 1$
  - convex functions have $\varphi'' \geq 0$ everywhere (slope is increasing)
    - x^2, x^4, … (x^n for any even n)
    - $|x|$
    - a^x (for $a > 1$)
    - $1/x$ (for $x > 0$)
    - $\max(x_1, x_2, \ldots)$
  - concave functions have $\varphi'' \leq 0$ everywhere (slope is decreasing)
    - $\sqrt{x}$
    - $\log x$
    - x^p for $0 < p < 1$
    - $\min(x_1, x_2, \ldots)$
  - linear functions are both convex and concave
  - worth memorizing: $\log \mathbb{E}[X] \geq \mathbb{E}[\log X]$

- **AM-GM inequality** (arithmetic mean, geometric mean): arithmetic mean always $\geq$ geometric mean

$$\frac{a_1 + \cdots a_n}{n} \leq \sqrt[n]{a_1 \cdots a_n}$$

  - proof: take log of both sides, apply Jensen's inequality (using the fact that log is concave), then exponentiate both sides
- **Cauchy-Schwarz inequality**

$$\mathbb{E}[XY]^2 \leq \mathbb{E}[X^2]\mathbb{E}[Y^2]$$

  - another form

$$\left(\sum_{i=1}^{n} x_i y_i\right)^2 \leq \left(\sum_{i=1}^{n} x_i^2\right)\left(\sum_{i=1}^{n} y_i^2\right)$$

- **union bound**

$$P(A_1 \cup A_2 \cup \cdots \cup A_n) \leq P(A_1) + P(A_2) + \cdots + P(A_n)$$

Conditional probabilities

- suppose we have two random variables, X and Y
  - **joint distribution**: $P(X = x, Y = y)$
  - **marginal distribution**: $P(X = x) = \sum_y P(X = x, Y = y)$
  - **conditional distribution**: $P(X = x \mid Y = y)$
- **Bayes rule**

$$P(A \mid B) = \frac{P(B \mid A)P(A)}{P(B)}$$

  - $P(A)$ is the **prior** (what we believed before seeing B)
  - $P(A \mid B)$ is the **posterior** (what we believed after seeing B)
  - calculate $P(B)$ with law of total probability
- **law of total expectation** (also known as the **tower property**)

$$\mathbb{E}[X] = \sum_y \mathbb{E}[X \mid Y = y] \cdot P(Y = y) \quad \text{or} \quad \int_{-\infty}^{\infty} \mathbb{E}[X \mid Y = y] f(y)\, dy$$

  - can also be stated as $\mathbb{E}[X] = \mathbb{E}_Y[\mathbb{E}[X \mid Y]]$
    - note that in this statement, $\mathbb{E}[X \mid Y]$ is itself RV expressed in terms of Y
- **law of total probability**

$$P(A) = \sum_y P(A \mid Y = y) \cdot P(Y = y)$$

Limit theorems

- **central limit theorem**: mean of n iid RVs X_i with mean μ and variance σ^2 is Gaussian, no matter what the X_i look like

$$\bar{X}_n = \frac{1}{n}\sum_{i=1}^{n} X_i \approx \mathcal{N}\left(\mu, \frac{\sigma^2}{n}\right)$$

  - derivation of variance

$$\mathrm{Var}(\bar{X}_n) = \mathrm{Var}\left(\frac{1}{n}\sum_{i=1}^{n} X_i\right) = \frac{1}{n^2}\mathrm{Var}\left(\sum_{i=1}^{n} X_i\right) = \frac{1}{n^2}\sum_{i=1}^{n}\mathrm{Var}(X_i) = \frac{n}{n^2}\sigma^2 = \frac{\sigma^2}{n}$$

  - **standard error** is the standard deviation of $\bar{X}_n$, which is $\sigma/\sqrt{n}$
    - what it means: error shrinks like $\sqrt{n}$, need 100x samples to reduce error 10x
  - $(1-\alpha)$ confidence interval is given by

$$\bar{X}_n \pm z_{1-\alpha/2} \cdot \frac{\sigma}{\sqrt{n}}$$

- **law of large numbers**: mean of n iid RVs X_i with mean μ converges to the true average

$$P(|\bar{X}_n - \mu| > \epsilon) \to 0 \quad \text{for any } \epsilon > 0$$

  - proof from Chebyshev's (substitute $\epsilon = k\sigma$)

$$P(|\bar{X}_n - \mu| > \epsilon) \leq \frac{\mathrm{Var}(\bar{X}_n)}{\epsilon^2} = \frac{\sigma^2}{n\epsilon^2} \to 0$$

Some problem types

First-step analysis

- what's the expected number of coin flips to get two heads in a row?
  - let
    - E_0 = expected flips to finish after just flipping tails (or just started)
    - E_1 = expected flips after just flipping heads

$$E_0 = \underbrace{\frac{1}{2}(1 + E_0)}_{\text{tails, no progress}} + \underbrace{\frac{1}{2}(1 + E_1)}_{\text{heads}}$$

$$E_1 = \underbrace{\frac{1}{2}(1 + E_0)}_{\text{tails, start over}} + \underbrace{\frac{1}{2} \cdot 0}_{\text{heads again}}$$

  - then solve system of linear equations to get $E_0 = 6$

> first-step analysis trick: focus on the first step of the process, and express the expectations we want to solve for in terms of themselves. then solve system of linear equations.

Counting in expectation

- how many things are expected to satisfy some property?
  - define $X_i = 1[\text{item } i \text{ has property}]$
  - define a sum of indicator random variables $N = \sum_i X_i$
  - compute $\mathbb{E}[X_i] = P(X_i)$ for each i (use symmetry)
  - apply linearity of expectation: $\mathbb{E}[N] = \sum_i P(X_i)$
- examples: expected number of fixed points in a random permutation
  - by symmetry, position i equally likely to have any item, so $P(X_i = 1) = 1/n$

Max/min of n RVs

- if $X_1, \dots, X_n$ are iid with common CDF F_X
  - for the max $M = \max_i X_i$,

$$F_M(x) = P(M \le x) = P(\text{all } X_i \le x) = [F_X(x)]^n$$

  - for the min $m = \min_i X_i$,

$$P(m > x) = P(\text{every } X_i > m) = [1 - F_X(x)]^n$$
$$F_m(x) = 1 - P(m > x)$$

- two different approaches to calculate expectation of min or max
  - 3-step recipe: CDF → PDF → integrate
  - for non-negative RVs, can use tail formula directly $\mathbb{E}[X] = \int_0^\infty P(X > x)\,dx$

- let's do this for $X_i \sim \text{Uniform}[0,1]$
  - max $M = \max_i X_i$
    - CDF: $F_M(x) = \prod_{i=1}^n P(X_i \le x) = x^n$
    - PDF: $f_M(x) = \frac{d}{dx} x^n = nx^{n-1}$
    - expectation: $\mathbb{E}[M] = \int_0^1 x \cdot nx^{n-1}\,dx = n\int_0^1 x^n\,dx = \frac{n}{n+1}$
    - intuition: n uniform points X_i split the segment into $n+1$ segments which are all equivalently distributed (by symmetry), so each must have mean length $1/(n+1)$
  - alternatively with tail formula: $\mathbb{E}[M] = \int_0^1 (1-x^n)\,dx = x - \frac{x^{n+1}}{n+1}|_0^1 = \frac{n}{n+1}$
  - min $m = \min_i X_i$
    - $P(m > x) = (1-x)^n$
    - CDF: $F_m(x) = P(m \le x) = 1 - P(m \ge x) = 1-(1-x)^n$
    - PDF: $f_m(x) = \frac{d}{dx} F_m(x) = n(1-x)^{n-1}$
    - expectation: $\mathbb{E}[m] = \int_0^1 x \cdot n(1-x)^{n-1}\,dx = \frac{1}{n+1}$
- for $X_i \sim \text{Exponential}(\lambda)$
  - min
    - $P(m > x) = [P(X_i > x)]^n = e^{-n\lambda x}$
    - $\mathbb{E}[m] = \int_0^\infty e^{-n\lambda x}\,dx = \frac{1}{n\lambda}$
  - max is a lot more complicated, but good to know that it grows logarithmically with n
- for $X_i \sim \mathcal{N}(0,1)$, $\mathbb{E}[\max_i X_i] \approx \sqrt{2\ln n}$ (this is pretty difficult to derive)

Combinatorics

- **combination**: choosing k items from n, order doesn't matter

$$\binom{n}{k} = \frac{n!}{k!(n-k)!} = \frac{P(n,k)}{k!}$$

  - the relation to permutation: every set of k items can be arranged in $k!$ orders, and combinations collapse the orderings into one
- **permutation**: arranging k items from n, order matters

$$P(n,k) = \frac{n!}{(n-k)!}$$

- number of ways to arrange n objects when there are n_1 of type 1, n_2 of type 2, …

$$\binom{n}{n_1, \ldots, n_m} = \frac{n!}{n_1! \cdots n_m!}$$

  - permute all n objects, the divide out the rearrangements of items within each type
- a **derangement** is a permutation with no fixed points

- **combination with replacement**: choosing k items from n types with replacement

$$\binom{n+k-1}{k}$$

  - this is the **stars and bars** argument: each choice can be represented as a sequence of stars and bars, where there are k stars and $n-1$ bars → this is choosing k placements of stars from $n+k-1$ positions
- how many ways are there to write non-negative integer n has sum of k non-negative integers?
  - visualize n stars with $k-1$ bars among them

order matters?	replacement?	count
yes	no	$\frac{n!}{(n-k)!}$ (permutations)
no	no	$\binom{n}{k}$ (combinations)
yes	yes	n^k
no	yes	$\binom{n+k-1}{k}$ (stars and bars)

- **inclusion-exclusion** principle

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$
$$P(A \cup B \cup C) = P(A) + P(B) + P(C) - P(A \cap B) - P(A \cap C) - P(B \cap C) + P(A \cap B \cap C)$$

  - as counts

$$|A \cup B \cup C| = |A| + |B| + |C| - |A \cap B| - |A \cap C| - |B \cap C| + |A \cap B \cap C|$$

Markov chains

- **Markov chain** is a sequence of RVs $X_0, X_1, X_2, \ldots$ taking values in some state space S satisfying the **Markov property**: the future depends only on the present
- for finite state spaces, the chain is described by the **transition matrix** P where $P_{ij} = P(X_{t+1} = j \mid X_t = i)$
  - rows sum to 1 and are probability distributions over next-states
- if π_t is the distribution over states at time t, then $\pi_{t+1} = \pi_t P$
- a **stationary distribution** satisfies $\pi = \pi P$
  - under mild conditions, the chain has a unique stationary distribution and $\pi_t \to \pi$ regardless of where we started

- the expected return time is the reciprocal of its stationary probability

$$\mathbb{E}[\text{return to state } i] = \frac{1}{\pi_i}$$

- general principle to solve for expected absorption time
  - use first-step analysis to express the recurrence
    - expected time h_j to absorption from state j
      - $h_A = 0$ for target absorbing state A

$$h_i = 1 + \sum_j P_{ij} h_j$$

    - probability of absorption q_j from state j
      - $q_A = 1$ for desired absorbing state A, and $q_B = 0$ for all other absorbing states B

$$q_i = \sum_j P_{ij} q_j$$

  - solve recurrence
    - for small chains, can just solve the linear system
    - for chains with structure, guess a closed form and verify
      - to make an educated guess, compute the answer for small cases and try to generalize
      - think about what makes sense in terms of function shape
    - fit boundary conditions

- handy techniques
  - symmetry can allow you to collapse equivalent states
    - for a random walk on a circle, easier to reparameterize h_i to be expected hitting time from a particular distance rather than from a particular state
  - use stationary distribution to provide shortcuts for expected hitting time
    - on a graph where every node looks the same (cycles, complete graphs, etc.), expected hitting time h between adjacent vertices is always $n-1$ where n is the number of nodes
      - this is because $\mathbb{E}[\text{return to self}] = n$ (since stationary distribution has $1/n$ everywhere)
      - and $\mathbb{E}[\text{return to self}] = 1 + h$ because each of the d adjacent nodes contributes $\frac{1}{d}h$

Puzzles

Birthday problem

- how many people n does it take so that the probability of two people having the same birthday exceeds 50%?
- $P(\text{collision}) = 1 - P(\text{all unique birthdays})$
- let d be number of days in the year and n the number of people

$$P(\text{all unique birthdays}) = \frac{d}{d} \cdot \frac{d-1}{d} \cdot \frac{d-2}{d} \cdots \frac{d-n+1}{d} = \frac{d!}{(d-n)! \cdot d^n}$$

Gambler's ruin

- you start with \$$a$ dollars
  - win \$1 with probability p and lose \$1 with probability $1-p$
  - stop when you reach \$$N$ (you win) or \$0 (you're ruined)
  - probability you reach \$$N$?
- let P_i be probability of reaching N starting from i
  - boundary conditions: $P_0 = 0, P_N = 1$
- for any starting amount $0 < i < N$,

$$q_i = p \cdot q_{i+1} + (1-p) \cdot q_{i-1}$$

  - second-order linear recurrence

- when the game is fair, the probability of hitting N when starting at a is simply a/N
  - approach 1: express expected value at stopping time in two different ways
    - let T be the stop time, and W_T be the wealth at time T
    - we know $\mathbb{E}[W_T] = a$ (no drift)
    - we also know W_T is either 0 or N, so

$$\mathbb{E}[W_T] = q_a \cdot N + (1 - q_a) \cdot 0 = q_a \cdot N$$

    - combining these, we get $q_a = a/N$
  - approach 2: solve recurrence

$$q_i = \frac{1}{2} q_{i-1} + \frac{1}{2} q_{i+1} \implies q_{i+1} - q_i = q_i - q_{i-1}$$

    - all consecutive differences are equal → q_i is a linear function of i
    - with $q_i = 0$ and $q_N = 1$, we get $q_i = i/N$
- this is a random walk with absorbing barriers at 0 and N
- in general, a **martingale** is a sequence of RVs $M_1, \ldots, M_n$ such that $\mathbb{E}[M_1] = \cdots = \mathbb{E}[M_n]$
- the expected absorption time h_i is $i(N - i)$
  - this is tricky to derive, but easy to verify by making sure it satisfies the recursive relationships given by first-step analysis

Random walk

- simple symmetric **random walk**: let $S_n = X_1 + \cdots + X_n$ where each $X_i = \pm 1$ with equal probability

- obviously $\mathbb{E}[S_n] = \mathbb{E}[X_i] = 0$
- each $X_i = \pm 1$ has variance $\mathbb{E}[X_i^2] - (\mathbb{E}[X_i])^2 = 1 - 0 = 1$

$$\mathrm{Var}(S_n) = \sum_{i=1}^{n} \mathrm{Var}(X_i) = n$$

  - typical distance from origin (generally measured by standard deviation) grows like $\sqrt{n}$
- in the biased case ($X_i = 1$ with p, else $X_i = -1$)

$$\mathbb{E}[S_n] = n(p \cdot 1 + (1 - p) \cdot (-1)) = n(2p - 1)$$

Coupon collector

- n different types of coupons. every time you buy a cereal box, you get one coupon uniformly at random. in expectation, how much time does it take to collect all n coupons?

- let phase k denote the phase in which we have k types
  - each phase $X_k \sim \text{Geom}((n-k)/n)$
  - the expected length of phase k is

$$\mathbb{E}[X_k] = \frac{n}{n-k}$$

  - total expected time

$$\mathbb{E}[X] = \sum_{k=0}^{n-1} \mathbb{E}[X_k] = \sum_{k=0}^{n-1} \frac{n}{n-k} = n \sum_{j=1}^{n} \frac{1}{j} = n \cdot H_n$$

    - where $H_n = 1 + 1/2 + 1/3 + \cdots + 1/n$ is the n-th **harmonic number**
  - final phase takes n boxes in expectation!

Information-theoretic puzzles

- count the information
- if we need to distinguish k possibilities and each test has m outcomes, then we need at least $\lceil \log_m k \rceil$ tests
  - this is a lower bound
  - whether tests exist to achieve this bound depends on the constraints of the problem

- 9 coins, one is heavier, 2 weighings
  - divide into 3 groups of 3
    - left side heavy → counterfeit on left side
    - right side heavy → counterfeit on right side
    - balanced → counterfeit in the left-out set
  - then weigh 2 of the remaining 3

Reservoir sampling

- we receive a stream of items: $x_1, x_2, x_3, \ldots$ at the end of the stream, return a single item uniformly at random from all the items seen (length n of stream is not known ahead of time)
- the algorithm
  - keep the first item: $s = x_1$
  - for $n = 2, 3, 4, \ldots$
    - with probability $1/n$, replace s with x_n

- proof of correctness by induction
  - claim: after processing n items, each of $x_1, \ldots, x_n$ is stored in s with probability $1/n$
  - base case ($n = 1$): trivially, $s = x_1$ with probability $1 = 1/1$
  - inductive step: after processing x_n, there are two possible outcomes
    - $s = x_n$ (replace): this happens with probability $1/n$
    - $s = x_k$ for some $k < n$ (keep): this requires (1) $s = x_k$ after step $n - 1$, which has probability $1/(n-1)$ by induction, and that (2) we rejected x_n, which has probability $(n-1)/n$

$$\frac{1}{n-1} \cdot \frac{n-1}{n} = \frac{1}{n}$$

  - thus every x_k is in the reservoir with probability $1/n$ after n steps

Secretary problem

- problem: n candidates interviewed in random order, after each interview, decide immediately to hire or reject that candidate. our goal is to maximize probability of hiring the best candidate. we use the strategy: reject the first k candidates, then hire the next candidate better than all the first k (if no such candidate appears, hire the last one). what is the optimal k?
- let $P(k) =$ probability that strategy with threshold k hires the best candidate, let b be the position of the best candidate
  - for the strategy to succeed:
    1. $b > k$: best candidate appears after the first k
    2. the best candidate among positions $1, \ldots, b-1$ is in $1, \ldots, k$ (otherwise we hire someone before getting to b)

$$\begin{aligned}
P(k) &= \sum_{b=k+1}^{n} P(\text{best candidate at } b) P(\text{best among } 1, \ldots, b-1 \text{ appears } \leq k) \\
&= \sum_{b=k+1}^{n} \frac{1}{n} \frac{k}{b-1} \\
&= \frac{k}{n} \sum_{b=k+1}^{n} \frac{1}{b-1} \\
&= \frac{k}{n} \sum_{i=k}^{n-1} \frac{1}{i} \\
&= \frac{k}{n} (H_{n-1} - H_{k-1}) \\
&\approx \frac{k}{n} (\log(n-1) - \log(k-1)) && H_n \approx \log n \text{ for large } n \\
&\approx \frac{k}{n} \log \frac{n}{k} \\
&= r \log \frac{1}{r} && \text{let } r = \frac{k}{n} \\
&= -r \log r
\end{aligned}$$

  - taking the derivative, we see $-\frac{d}{dr} r \log r = -\log r - 1$
    - setting this to 0, we get $\log r = -1 \implies r = \frac{1}{e} \implies k = \frac{n}{e}$

Monty Hall